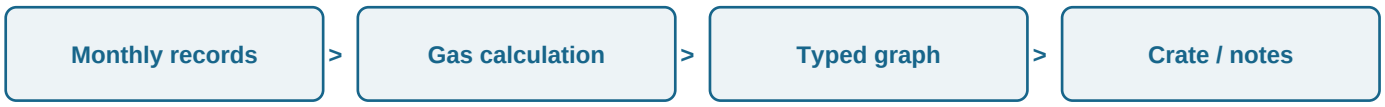
MOENV evidence: 2023 and 2024

Primary worked case | 16 October 2026

A fully fictional single-site manufacturing case, mapped to inspected Taiwan MOENV collection guidance. This is an evidence workflow, not a filing or assurance opinion.



Executed evidence	Observed
Monthly rows / documents	120 / 122
Result lineages	4 in 2023; 5 in 2024
CLI calls	33 (30 success, 3 expected rejection)

INTERPRET / All nine results remain unreviewed. Historical electricity vintages and synthetic fuel LHV are explicit limitations.

NEXT / Install the locked environment.

01 / Prepare and execute

Run from the repository root.

Prerequisites: Git, uv, Python 3.12; tested with 3.12.14. Installation may download packages. Runtime needs no model, API key or paid service.

```
git clone https://github.com/simon5530/ghg-assurance-graph.git
cd ghg-assurance-graph
git rev-parse HEAD
uv sync --locked
uv run python scripts/run_moenv_example.py --out artifacts/moenv-worked
```

OUTPUT below means artifacts/moenv-worked. The destination must not exist. Open OUTPUT/REPORT.md, commands.json and summary.json. Repeat into a new folder, not over saved evidence.

INTERPRET / The runner invokes real CLI parsers and records stdout, stderr and exit codes. Its Python socket guard is not an OS sandbox.

NEXT / Follow the collection-to-evidence folders.

02 / Map the collection forms

Use original internal forms, not company disclosure totals.

Forms	Purpose
01-03	Entity, facility boundary, source/equipment register
04-05	Monthly activity and gas-level quantification
06-08	Quality, uncertainty, annual totals
09-11	Energy, production, tests and test dates
12-13	Project extensions: evidence index and change records

Read examples/moenv_training/FIELD_MAP.md and SOURCES.md. Official inputs span a 113-year guideline/template and a June-2026 workbook update; they are not one publication date.

The registration is deliberately non-real. A facility/registered boundary is not interchangeable with corporate operational-control metadata. Omitted sources follow this fictional equipment scenario only.

INTERPRET / Tables 12/13 are project traceability extensions, not official mandates. Unknown calibration and uncertainty are not fabricated pass scores.

NEXT / Inspect the monthly original documents.

03 / Inspect original monthly balances

Open generated/2023/04-activity.csv and evidence/.

Source	How quantity is derived
Gas / electricity	(Closing meter - opening meter) x multiplier
Diesel	Opening stock + purchases - closing stock
HFC-134a	Opening mass + added - recovered - closing mass
New generator	Not installed in 2023; starts July 2024

The January-2023 boiler row is 1010 m3; its synthetic meter starts at 100000 and closes at 101010. Quantity, period, unit and source must match the referenced file and SHA256.

A documented inactive zero is not a missing record. Refrigerant charge or purchases alone do not establish leaked mass. Real observations and calibration cannot be authenticated by synthetic examples.

INTERPRET / Monthly documents are originals of the training scenario, not real invoices. Preserve them alongside the CSV and source index.

NEXT / Run the collection-specific checker.

04 / Check inputs before the graph

The two checks have different contracts.

```
uv run python -m ghg_assurance_graph.moenv_example --check artifacts/moenv-
worked/generated/2024
uv run ghgag validate artifacts/moenv-worked/generated/2024/package.json
```

Check	Expected output
Collection	60 monthly rows and 60 document checks
Raw balances	Meter / fuel / refrigerant quantities reconcile
Canonical package	conforms: true; findings: []
Meaning	Selected rules only, not source truth

Collection checking rejects gaps, duplicates, blank values, wrong units/years, corrupted evidence and altered results. ghgag validate alone does not read the monthly CSV/documents.

INTERPRET / Never replace unknown quantities with zero to satisfy the checker. Both successful checks are needed for the documented workflow.

NEXT / Trace the gas calculation and its projection.

05 / Understand gas arithmetic

Read generated/2023/05-quantification.csv.

```
2023 boiler: 12780 m3 x 38 MJ/m3 / 1000000
x (56100 + 1 x 28 + 0.1 x 265)
= 27270.87138 kgCO2e
```

Parameter	Status
kg gas / TJ coefficients	Transcribed official stationary-fuel values
GWP	CO2=1, CH4=28, N2O=265, HFC-134a=1300
LHV 38 / 36 MJ per unit	Synthetic, not tested or official defaults
Electricity 0.494 / 0.474	Historical example-workbook vintage replay

The graph receives an explicit composite kgCO2e/unit factor. Do not multiply GWP again. Companion gas tables do not implement a gas-resolved core schema or prove applicable filing-factor selection.

INTERPRET / Official factors, synthetic LHV and historical-vintage policy remain distinct. Fuel category is stationary backup generation, not a vehicle factor.

NEXT / Check the annual source results.

06 / Read the annual totals

Compare the same declared fictional site.

Source (kgCO2e)	2023	2024
Natural-gas boiler	27270.87138	24710.22618
Existing diesel generator	1493.402184	1332.821304
New diesel generator	0 (not installed)	321.16176
HFC-134a leakage	3900.0	2600.0
Purchased electricity	63133.2	66265.2
Total	95797.4735640	95229.4092440

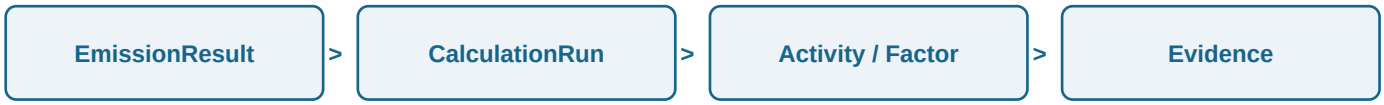
The 2024 new generator contributes 321.16176 kgCO2e. The site boundary is unchanged; equipment composition changes. Purchased electricity rises while boiler, existing generator and refrigerant quantities decline.

INTERPRET / The annual decrease is -568.064320 kgCO2e. It is not independently established abatement or a complete corporate inventory.

NEXT / Inspect actual graph explanations.

07 / Explain an emissions result

Use the exact revision URN and explicit package path.




```
uv run ghgag graph explain urn:ghgag:moenv-demo-2024-electricity-result:v1 --input
artifacts/moenv-worked/generated/2024/package.json
```

Returned field	Observed value / meaning
value / unit	66265.2 / kg_CO2e
status	unreviewed
activity / factor	Declared annual activity and coefficient records
evidence / factorEvidence	Different synthetic evidence bundles
period / boundary / review	Explicit references, not inferred acceptance

All 9 primary-case lineages resolve. Annual activity bundles contain 12 monthly evidence IDs and hashes; the actual documents are in generated/YEAR/evidence.

INTERPRET / A linked citation is not a downloaded or authenticated document. A retrospective review is not the calculation itself.

NEXT / Query factor usage and pending reviews.

08 / Keep review state visible

A clean graph must not become fictional acceptance.

```
uv run ghgag graph query unreviewed_results --input artifacts/moenv-
worked/generated/2024/package.json
uv run ghgag graph query results_using_factor --target urn:ghgag:moenv-
demo-2024-electricity-factor:v1 --input artifacts/moenv-
worked/generated/2024/package.json
```

Observed	Meaning
4 unreviewed / 2023	Every primary result still lacks human acceptance
5 unreviewed / 2024	New generator adds one result
50 / 61 canonical records	Typed record counts, not document counts
One electricity factor match	Exact-ID use, not arbitrary semantic search

INTERPRET / The approved query templates do not authenticate reviewers, run arbitrary SPARQL, or certify inventory completeness.

NEXT / Read the labelled arithmetic comparison.

09 / Compare without causal claims

Open OUTPUT/comparison.json.

Source	Delta kgCO2e
boiler	-2560.645200
cooling	-1300.000000
electricity	3132.000000
generator	-160.580880
new-generator	321.161760

The source changes sum to -568.064320 kgCO2e. Every causal explanation stays UNKNOWN. A zero predecessor is justified only for the explicitly uninstalled new generator.

Do not call ghgag diff on these MOENV folders as a supported main step. The legacy raw-snapshot contract is different and the negative control observes exit 2. Old benchmark component scores are not primary-case results.

INTERPRET / This is an arithmetic series comparison, explicitly not CarbonDiff or a measured emissions-reduction decomposition.

NEXT / Preserve canonical packages and original documents together.

10 / Package and verify

One-click output already contains verified packages/2023 and packages/2024.

```
uv run ghgag package verify artifacts/moenv-worked/packages/2024
```

Inside closed canonical crate	Outside, preserve with it
Canonical package and RDF exports	generated/YEAR/04-activity.csv
Schema / ontology / metadata	generated/YEAR/evidence/*
Manifest and software versions	Source ledger and collection forms

Both package verifications return valid: true. Use a new destination when creating another package. Creation timestamp 2026-10-16T00:00:00Z is an explicit replay input, not an authenticated review date.

Copy the complete OUTPUT directory to preserve the monthly chain; the seven-file canonical crate alone is not the complete source collection.

INTERPRET / Digests prove bounded consistency, not document authenticity. Keep the manifest digest independently; verification must not fetch citation URLs.

NEXT / Inspect graph-derived Markdown if useful.

11 / Export and inspect notes

No Obsidian installation is required for generation.

```
uv run ghgag export obsidian artifacts/moenv-2024-vault --input artifacts/moenv-
worked/generated/2024/package.json
```

Year	Actual note count
2023	127
2024	140

The example command needs a new destination. In the one-click output, start at vaults/YEAR/index.md. All generated wikilinks resolve as files; desktop Obsidian interaction remains untested.

Keep personal annotations separately. Exports are regenerated views, not an approval workflow, editable source of truth or general database.

INTERPRET / Textual notes expose the supplied graph; they do not establish an actual meter observation or performed human review.

NEXT / Verify the expected failure receipts.

12 / Inspect intentional failures

The runner records expected rejections, not false passes.

Separate negative control	Observed exit
Missing monthly activity row	2: collection request rejected
Extra file in canonical crate	2: invalid closed package request
Legacy ghgag diff on MOENV folders	2: unsupported input contract

Read negative/*.stderr.txt with commands.json. Unexpected failures stop the run and do not produce a success report. Valid source collections and packages stay untouched.

If a digest differs, investigate version/source changes. Do not patch only the manifest to claim true observations. If output exists, choose a new folder; do not overwrite evidence.

INTERPRET / The controls demonstrate selected software boundaries; three failures are not a defect-detection benchmark or a security certification.

NEXT / Understand how edits and handoff are bounded.

13 / Edit carefully and hand off

The workbook is not an automatic Excel-to-graph importer.

You may edit blue monthly cells in the filled XLSX examples to observe formula recalculation. Fixed source/month blocks must not move. Edits do not rewrite CSV, originals, hashes, canonical results or graphs.

--out regenerates the fixed training scenario; it does not read your edited workbook. --check reads this profile's saved CSV and rejects inconsistent documents or calculations. Arbitrary real-company ingestion requires an explicit mapping, not fictional evidence.

Handoff item	Check
Source commit / uv.lock	Record exact revision and dependency set
Two totals	95797.473564 / 95229.409244 kgCO2e
Receipts / lineages / originals	33 calls / 9 lineages / 122 synthetic documents
Review / uncertainty	Unreviewed / not quantified
Primary guide and paper	MOENV case, not old regression metrics

INTERPRET / No source truth, statutory conformity, independent assurance, complete Scope 3 coverage or causal abatement is claimed.

NEXT / Use docs/USAGE.zh-TW.md and the MOENV field/source ledgers for detailed commands.